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Subject : Programming

Subject code : PROG5121

Part 1 — Registration and login feature

```
/**
 * Handles user registration validation and login status.
 */
public class Login {

    private final String username;
    private final String password;
    private final String cellPhoneNumber;
    private final String firstName;
    private final String lastName;

    public Login(String username,
                 String password,
                 String cellPhoneNumber,
                 String firstName,
                 String lastName) {
        this.username = username;
        this.password = password;
        this.cellPhoneNumber = cellPhoneNumber;
        this.firstName = firstName;
        this.lastName = lastName;
    }

    /** Checks whether the username contains an underscore and is at most five
    characters. */
    public boolean checkUserName() {
        return username != null
            && username.contains("_")
            && username.length() <= 5;
    }

    /** Checks whether the password meets the required complexity rules. */
    public boolean checkPasswordComplexity() {
        return password != null
            && password.length() >= 8
            && password.matches(".*[A-Z].*")
            && password.matches(".*[0-9].*")
            && password.matches(".*[^a-zA-Z0-9].*");
    }

    /** Checks for a South African number in the +27 followed by nine digits
    format. */
    public boolean checkCellPhoneNumber() {
        return cellPhoneNumber != null
            && cellPhoneNumber.matches("\\+27\\d{9}");
    }
}
```

```

    }

    /** Returns the appropriate registration result message. */
    public String registerUser() {
        if (!checkUserName()) {
            return "Username is not correctly formatted, please ensure that
your "
                + "username contains an underscore and is no more than
five characters in length.";
        }

        if (!checkPasswordComplexity()) {
            return "Password is not correctly formatted; please ensure that
the password "
                + "contains at least eight characters, a capital letter, a
number, and a special character.";
        }

        if (!checkCellPhoneNumber()) {
            return "Cell phone number is not correctly formatted; please
ensure that the number "
                + "contains the international code and is no more than ten
characters in length.";
        }

        return "User has been registered successfully.";
    }

    /** Checks whether the entered credentials match the registered details.
*/
    public boolean loginUser(String enteredUsername, String enteredPassword) {
        return enteredUsername != null
            && enteredPassword != null
            && enteredUsername.equals(username)
            && enteredPassword.equals(password);
    }

    /** Returns the appropriate login status message. */
    public String returnLoginStatus(boolean loginStatus) {
        if (loginStatus) {
            return "Welcome " + firstName + " " + lastName
                + ", it is great to see you again.";
        }

        return "Username or password incorrect, please try again.";
    }

    public static void main(String[] args) {

```

```
        Login user = new Login("kyl_1", "Password1!", "+27821234567", "Kylie",  
"Smith");  
  
        System.out.println(user.registerUser());  
  
        boolean loginStatus = user.loginUser("kyl_1", "Password1!");  
        System.out.println(user.returnLoginStatus(loginStatus));  
    }  
}
```